

concepts

Glossary of Key Terms & Concepts

This document explains the key scientific concepts, data files, and analysis techniques used to create the lunar elemental abundance map.

Core Scientific Concepts

Solar Angle (`solarang_deg`)

- **Simple Definition:** The angle of the Sun in the sky as seen from a specific point on the lunar surface.
- **Explanation:** This is like the time of day. A solar angle of 0° means the Sun is directly overhead (lunar noon), providing the most intense illumination. An angle of 90° means the Sun is on the horizon (lunar sunrise/sunset). An angle greater than 90° means it's nighttime.
- **Role in Project:** This was our most important filter. We only used data with **low solar angles** because X-ray fluorescence is powered by sunlight.

X-ray Fluorescence (XRF)

- **Simple Definition:** The process where an atom absorbs a high-energy X-ray and then emits its own lower-energy X-ray with a unique "fingerprint" energy.
- **Analogy:** Imagine tapping a crystal glass. It always rings with a specific, pure note. Similarly, when a solar X-ray "taps" a silicon atom on the Moon, the silicon atom "rings" by emitting its own X-ray at a specific energy (~ 1.74 keV). By listening for these "notes," we can identify the "glasses" (the elements).
- **Role in Project:** This is the **fundamental physical process** we are measuring. The peaks we fit in the CLASS spectrum (Mg, Al, Si) are the XRF "fingerprints" from the lunar surface.

Regolith

- **Simple Definition:** The layer of loose dust, soil, and broken rock covering the Moon's solid surface.

- **Explanation:** This is the material that the solar X-rays interact with to produce the fluorescence signal that CLASS measures. Its composition is what we are mapping.

Spectrum

- **Simple Definition:** A graph that shows the intensity of light (or radiation) at different energies.
- **Explanation:** In our project, a spectrum is a histogram that shows the number of X-ray photons (**Counts**) detected in different energy bins (**Channels**). The peaks in the spectrum reveal the presence of specific elements.
- **Role in Project:** The spectra from CLASS (lunar) and XSM (solar) are the **primary data** we analyzed.

keV (kilo-electron Volt)

- **Simple Definition:** The standard unit of energy used for X-rays.
- **Explanation:** Just as we measure distance in meters, we measure the energy of high-energy photons like X-rays in electron Volts. A kilo-electron Volt is 1,000 electron Volts. The XRF fingerprints of elements like Mg, Al, and Si are all in the 1-2 keV range.

Data and Calibration Files

FITS (Flexible Image Transport System)

- **Simple Definition:** The universal file format for astronomical data, like a ZIP file for science.
- **Explanation:** A FITS file can contain multiple components, including image data, data tables, and text headers with metadata. Both CLASS and XSM data are stored in this format.

PHA (**.pha**) file

- **Simple Definition:** A standard FITS file structured to hold a spectrum.
- **Explanation:** It's a specific "flavor" of FITS file that software like **XSPEC** is designed to read. It contains the **COUNTS** per **CHANNEL** and has a header with essential information like **EXPOSURE** time.

- **Role in Project:** We created `.pha` files for the XSM data using `prepare_xsm_for_xspec.py` so they could be analyzed in `XSPEC`.

RMF (Response Matrix File)

- **Simple Definition:** A calibration file that describes the "blurriness" of an X-ray detector.
- **Analogy:** Think of it as the specification sheet for a blurry camera lens. It tells you exactly how a single, sharp point of light gets spread out into a fuzzy circle. In our case, it describes how a photon of a single, perfect energy gets spread across several detector channels.
- **Role in Project:** The RMF was essential for the **forward-fitting** process in `XSPEC`, allowing our model to simulate the instrumental blur.

ARF (Ancillary Response File)

- **Simple Definition:** A calibration file that describes the sensitivity or efficiency of an X-ray telescope.
- **Analogy:** This is like the "light-gathering power" of a telescope. It describes how good the instrument is at detecting photons of different energies. A telescope might be very sensitive to 2 keV photons but less sensitive to 10 keV photons.
- **Role in Project:** The ARF was used in both the `XSPEC` fit (as part of the `.rsp` file) and in our Python continuum subtraction script to correctly model the expected number of counts.

RSP (`.rsp`) file

- **Simple Definition:** A convenient file that combines both the RMF and ARF information into a single response file.
- **Role in Project:** The XSM calibration data provided a `.rsp` file, which we loaded into `XSPEC` to account for both the instrument's blur and its sensitivity at the same time.

Background File (`...bkg.pha`)

- **Simple Definition:** A pre-made spectrum of the instrument's own internal noise.

- **Explanation:** Every sensitive detector has a background "hum" or "static" from cosmic rays and its own electronics. This file is created by pointing the instrument at a blank source (like the dark side of the Moon) and recording that noise.
 - **Role in Project:** We subtracted this background template from both our XSM and CLASS data to help isolate the true signal.
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Analysis Techniques

Forward-Fitting (or Forward-Folding)

- **Simple Definition:** The standard analysis method in astrophysics where, instead of trying to "un-blur" your data, you create a perfect theoretical model and then "blur" it to match your messy real-world data.
- **Analogy:** You have a blurry photo of a license plate. Instead of trying to de-blur the photo (which can create artifacts), you take a guess at the number ("ABC-123"). You then take a picture of your guess with the *same blurry camera*. You compare the two blurry photos. You keep changing your guess ("ABC-124", etc.) until your simulated blurry photo perfectly matches the original one. Your final, correct guess is the answer.
- **Role in Project:** This was the core method used in `pyxspec` to analyze the XSM data. We built a physical model of the Sun (`apec`), and `XSPEC` folded it through the RMF/ARF to find the true solar temperature that best matched the data.

Unfolding (or Deconvolution)

- **Simple Definition:** The mathematical process of removing the known instrumental effects (the "blur" and "filter") from the observed data to estimate the true, original signal.
- **Role in Project:** This is the *result* of our forward-fitting process on the XSM data. The final output, `F_sun(E)` , is the unfolded solar spectrum.

Continuum

- **Simple Definition:** The smooth, underlying baseline of a spectrum, on top of which the sharp emission peaks sit.

- **Explanation:** In our CLASS data, the continuum is primarily caused by solar X-rays scattering off the lunar surface.
- **Role in Project:** The `run_continuum_subtraction` function was dedicated to modeling and subtracting this continuum to make the faint XRF peaks visible.

Gaussian Fit

- **Simple Definition:** A method to measure the properties of a peak by fitting a "bell curve" (a Gaussian function) to it.
- **Explanation:** The "blur" of the detector (described by the RMF) causes the sharp XRF fingerprints to be recorded as bell-shaped peaks. By fitting a Gaussian to these peaks, we can precisely measure their position, height, and width.
- **Role in Project:** The `py_fit_xrf_lines.py` script used a multi-Gaussian fit to measure the strength of the Mg, Al, and Si peaks in the final, clean residual spectrum.

Line Flux (Area)

- **Simple Definition:** The total strength of an emission line.
- **Explanation:** While a peak has a height, the more important physical quantity is its **area**. The area under a fitted Gaussian peak is proportional to the total number of photons from that element.
- **Role in Project:** This was the final scientific measurement we extracted. We calculated the area of the Mg, Al, and Si peaks to determine their relative abundance.

Mapping and Visualization

Footprint

- **Simple Definition:** The specific area on the Moon's surface that the CLASS detector was looking at during a single observation.
- **Explanation:** The `v0..v3` coordinates in the metadata define the four corners of this square-like footprint on the Moon.
- **Role in Project:** We used the coordinates of the footprint (both the center and the corners) to place our measurements on the final map.

Gridding / Binning

- **Simple Definition:** The process of creating a map by dividing a surface into a grid and averaging all data points that fall into each grid cell.
- **Role in Project:** Our final mapping script, `plot_healpix_map.py`, used this technique to convert the many scattered data points into a continuous map of the Moon. This method also correctly handles overlapping footprints by averaging them.

HEALPix

- **Simple Definition:** A professional method for dividing a sphere into pixels that all have the exact same surface area.
- **Analogy:** Imagine trying to wrap a soccer ball with square stickers. You'll get ugly, wrinkled bunches at the top and bottom. HEALPix is like a special set of diamond-shaped stickers that fit together perfectly to cover the ball with no wrinkles or stretching.
- **Role in Project:** We used the `healpy` library to create our grid. This ensures that our final map is scientifically "fair" and doesn't distort the areas near the lunar poles.